THE SCHEME THEORETIC IMAGE OF A MORPHISM FROM A REDUCED SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement has two numbered assertions; the proof handles them in two sentences with the relevant inputs named.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The universal property of the reduced induced structure on a closed subset. Properties of Spaces, Lemma spaces-properties-lemma-map-into-reduction, used at proof lines 3299-3302. Identifies the scheme theoretic image with the reduced induced structure.
- Preservation of reducedness under etale localization. used at proof lines 3303-3304. Carries the hypothesis to the base change along V -> Y.
- Openness of |V| -> |Y| for an etale morphism V -> Y. used at proof lines 3303-3304. Makes the closure computation compatible with the base change.

Source reference. Original lemma 03JJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4641–4641. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume X is reduced. Then*

(1) *the scheme theoretic image Z of f is the reduced induced algebraic space structure on $\overline{|f|(|X|)}$, and*
(2) *for any étale morphism $V \to Y$ the scheme theoretic image of $X \times_Y V \to V$ is equal to $Z \times_Y V$.*

Proof. Part (1) is true because the reduced induced algebraic space structure on $\overline{|f|(|X|)}$ is the smallest closed subspace of Y through which f factors, see Properties of Spaces, Lemma 03JJ. Part (2) follows from (1), the fact that $|V| \to |Y|$ is open, and the fact that being reduced is preserved under étale localization. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.